A Fundamental Approach to Transformer Thermal Modeling—Part I: Theory and Equivalent Circuit

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***Abstract*—A simple equivalent circuit to represent the thermal heat flow equations for power transformers is presented. Key features are the use of a *current source analogy* to represent heat input due to losses, and a *nonlinear resistor analogy* to represent the effect of air or oil cooling convection currents. An interesting historical note concerns the fact that the foregoing effect was first quantified in 1817. It is shown that the idea of "exponential response" is not the best way to think of the dynamics of the situation. It is also shown that one can consider ambient temperature to be a variable input to the system, and that it is properly represented as an ideal voltage source. Physical experimental verification, on an in-service 250 MVA transformer, is described in a companion paper.**

***Index Terms*—Power transformer protection, temperature, thermal factors.**

I. Introduction

THERE is an increasing emphasis on keeping transformers in service longer than has been the practice in the past, for economic reasons. It has been shown [1], [2] that life extension of several years can save considerable expense, and is prudent, provided that monitoring of the transformer is properly done. One aspect of monitoring is "dissolved gas analysis" (DGA). Another aspect is temperature, especially the "hot spot" temperature, meaning that point adjacent to a winding where the measured (fiber optic sensor) or design temperature is hottest: the winding hottest spot.

The heating and cooling processes within a large power transformer and its cooling equipment are not simple. Nevertheless, engineers must make simplifying assumptions in order to "get the job done." The latest IEEE standard [3] recommends that a long-standing method of calculating hot spot temperature be continued: Clause 7 of [3]. At the same time, there is a more rigorous method suggested: Annex G of the same document. The first method uses an "exponential rise" (and fall) concept, one that is appealing because it so closely parallels the everyday observance of water heating and cooling on a stove, or in a coffee cup. In fact the two process are very similar. The other appeal is that many electrical engineers are very familiar with the charge-discharge process for an RC circuit, so there is an appealing analogy. The more rigorous method, due to Pierce [4], uses purely thermodynamics and heat transfer principles.

There are some relatively recent papers [5]–[8] elaborating on the use of the Clause 7 equations.

Manuscript received December 2, 1999.

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Publisher Item Identifier S 0885-8977(01)01548-5.

TABLE I
Thermal–Electrical Analogous Quantities [10]

	THERMAL	ELECTRICAL
THROUGH VARIABLE	heat transfer rate, q watts	current, i amps
ACROSS VARIABLE	temperature, θ degrees C*	voltage, v volts
DISSIPATION ELEMENT	thermal resistance, R_{th} degC/watt	elect. resistance, R_{el} ohms
STORAGE ELEMENT	thermal capacitance, C_{th} joules/degC	elect. capacitance, C_{el} farads

*An alternative to *degrees Celsius(C)* is *degrees kelvin(K)*.

In mechanical engineering, there is a discipline known as heat transfer. Heat transfer differs from thermodynamics in that the latter considers only states of equilibrium, whereas the former includes the time variable as well [9], an essential factor in this context.

In a nutshell, the objective here is to re-examine the source of the equations used for heat transfer calculations within a power transformer, and arrive at a model which is not only accurate, but easy to use for practical calculations.

II. Heat Transfer Principles

A. Thermal-Electrical Analogy

The two important thermal parameters of a substance such as cooling transformer oil are its heat capacity and its thermal conductivity [10]. These are general properties which can be made specific to a particular volume of oil, in which case they become thermal capacitance and thermal resistance.

The thermal-electrical analogy is defined in Table I.

The *electrical* laws governing resistance and capacitance are, of course

$$v = R_{el} \cdot i \quad \text{and} \quad i = C_{el} \cdot \frac{dv}{dt} \tag{1}$$

where the symbols are defined in Table I.

The corresponding thermal laws are

$$\theta = R_{th} \cdot q \quad \text{and} \quad q = C_{th} \cdot \frac{d\theta}{dt} \tag{2}$$

where again the symbols are defined in Table I.

For the heat transfer case, the thermal resistance may be nonlinear, as explained in the section to follow:

WTF? kaj je theta tempa ali temp diff?

$$\theta = R_{thR} \cdot q^n \tag{3}$$

where R_{thR} is the "rated" value of R_{th}, i.e., the value for a known set of θ, q and n.

It is interesting to note, parenthetically, that there is no thermal quantity analogous to electrical *inductance*.

B. Heat Transfer Through a Wall Next to a Moving Fluid

When heat is flowing through a wall, such as the steel tank wall of a transformer, the rate of flow is directly proportional to the temperature difference across the wall *provided* that the rate of flow of heating or cooling fluid alongside the wall is constant.

Consider the oil-wall-air situation at a transformer tank wall or a cooling tube wall, with no fans. Suppose the temperature difference across the wall is doubled. Then the heat flow through the wall doubles due to that cause, but is *more-than-doubled* because the hotter air next to the wall will be moving faster due to its lighter weight, i.e., the *convection will be greater*. The result of these two effects is that the proportionality is not linear:

$$q = 1/R_{thR} \cdot \theta^{1/n} \qquad \text{with } 1/n > 1.0, \tag{4}$$

where θ is the temperature difference. A typical value for $1/n$ is 1.25, that is, $n = 0.8$, for the case of no fans. If the air is forced to flow faster, by fans, then n rises toward 1.0 because the rate of convection is no longer dependent on temperature. There is another effect: the nominal thermal resistance R_{thR} becomes much lower when there are fans, that is, the cooling is more effective. In summary, R_{thR} is not only linear, but also lower, as compared to the case with no fans.

The exponent defining the nonlinearity is traditionally n if the moving fluid medium is air, and m if it is oil.

III. Application to a Transformer Model

A. Oil-to-Air Model

Fig. 1 shows the thermal model represented as an electrical equivalent circuit. Simpler models—such as an RC circuit with a voltage source and a switch—are not adequate, or indeed correct under all conditions of fans/no-fans or varying ambient temperature. ojej ojej !!!!

The oil thermal resistance R_{oil} is a combination of the oil resistance and the interface resistance. Note that the "losses" input is properly represented as two current (that is, heat) sources, not voltage sources. In other words, the copper and iron losses *define ideal heat sources*, not ideal temperature sources. Conversely, the ambient air *defines* an *ideal temperature source*, not an ideal heat source.

Note that there are three input sources: the iron loss, the copper loss and the ambient temperature. The ambient air, it is true, is usually at a temperature below that of the oil, but this doesn't affect the equations. An "ideal source" is mathematically identical to an "ideal sink."

For this "lumped capacitance" analogy to be valid, there are some conditions [11]. Applying those conditions here, we first assume that the temperature distribution within the medium (oil) is uniform at any given instant. We know that this isn't true, but there is a "test" to determine whether or not the assumption is reasonable [12]. We next calculate the *Biot number(Bi)*, defined as the ratio of the thermal oil resistance to the thermal air resistance. As long as this ratio is low (≪1) the assumption of lumped capacitance is reasonable. In our case this is so, because the oil has a much lower thermal resistance than does the air. (Incidentally, the steel tank wall is not a significant factor because it is essentially a perfect heat conductor.)

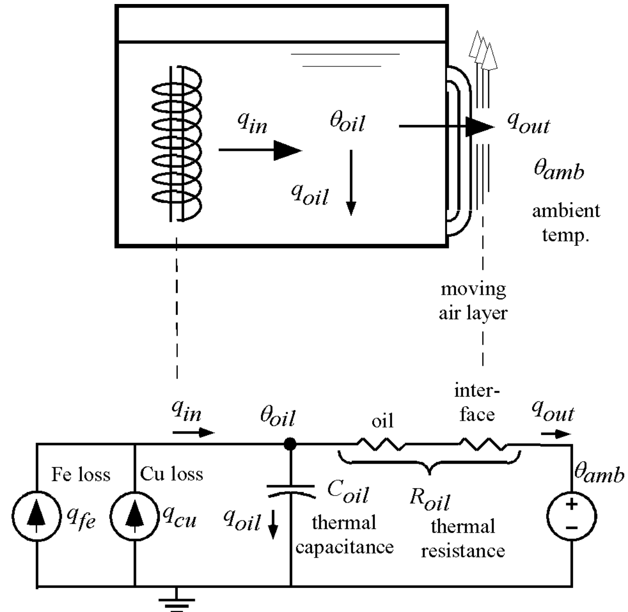

Fig. 1. Thermal model of oil-to-air heat transfer.

B. Differential Equations

The differential equation for the equivalent circuit shown in Fig. 1 is:

$$q_{fe} + q_{cu} = C_{oil} \cdot \frac{d\theta_{oil}}{dt} + \frac{1}{R_{oilR}} \cdot [\theta_{oil} - \theta_{amb}]^{1/n} \tag{5}$$

where q_{fe} is the heat generated by iron losses, q_{cu} is the heat generated by copper losses, C_{oil} is the thermal capacitance of the oil, θ_{oil} is the temperature of the oil, R_{oilR} is the thermal resistance of the oil under rated conditions, and θ_{amb} is the ambient temperature. See Fig. 1.

If we define

β = ratio of q_{cu} to q_{fe} at rated load ($I_{pu} = 1.00$),
τ_{oil} = $R_{oilR}C_{oil}$,
$\Delta\theta_{oil}$ = $\theta_{oil} - \theta_{amb}$, and
R as an appended subscript indicates rated load, steady-state, ambient temperature 30°C conditions,

then (5) reduces to

$$\frac{I_{pu}^2\beta + 1}{\beta + 1} \cdot [\Delta\theta_{oilR}]^{1/n} = \tau_{oil}\frac{d\theta_{oil}}{dt} + [\theta_{oil} - \theta_{amb}]^{1/n} \tag{6}$$

where $\Delta\theta_{oilR}$ is the rated load, rated ambient value of $\Delta\theta_{oil}$.

TABLE II
ANALOGY BETWEEN OIL-TO-AIR MODEL AND WINDING-TO-OIL MODEL

	OIL-TO-AIR	WINDING-TO-OIL
AMBIENT FLUID	AIR	OIL
FLUID MOVEMENT	BY FANS	BY PUMPS
CALCULATED TEMPERATURE	OIL	WINDING INSULATION
NONLINEARITY EXPONENT	n	m

The *difference equation* corresponding to (7) is

$$\mathbf{D}\theta_{oil} = \frac{\mathbf{D}t}{\tau_{oil}} \cdot \left[\frac{I_{pu}^2\beta + 1}{\beta + 1} \cdot [\Delta\theta_{oilR}]^{1/n} - [\theta_{oil} - \theta_{amb}]^{1/n}\right] \tag{7}$$

where $\mathbf{D}$ is the *difference* operator, indicating a small change in the associated variable. Solution of this equation for $\theta_{oil}(t)$ for arbitrary time functions of $I_{pu}(t)$ and $\theta_{amb}(t)$ is straightforward: each new value of $\mathbf{D}\theta_{oil}$ is added to the old value of θ_{oil} at each time step.

It is useful to summarize the symbols used in (6) and (7):

Independent variable: t

Input (forcing function) variables: I_{pu}, θ_{amb}

Output variable: θ_{oil}

Parameters (constants): β, $\Delta\theta_{oilR}$, τ_{oil}, n.

Note that the temperature that best represents the oil as a whole is traditionally taken to be the *top oil temperature*, that is, the temperature indicated by a nonelectrical gauge on the transformer tank wall, often labeled *liquid temperature*. There is an argument that the representative temperature would better be taken as the *bottom* oil temperature, but this is seldom a measurable quantity (if such is desired) on North American installed transformers.

In Annex G of the Guide [3] Pierce has shown that the "duct oil temperature" may be unusually high during a transient loading condition, especially if n is not equal to unity. This condition is not considered here, because of the difficulty in establishing the pertinent parameters.

Physical verification of this model, for a large power transformer with $n = 1$, is shown in a companion paper.

C. Winding-to-Oil Model

Table II shows how the *winding-to-oil* or *hot spot* model is analogous to the *oil-to-air* model.

Here, the "ambient" is the oil. The top oil temperature calculated from the oil-to-air model becomes the ambient temperature under which this thermal system operates. Note that the oil temperature affects the hot spot temperature, but not vice versa, which matches reality. A detail here is that it is not the *total* copper losses that generate the heat, but only, say 0.1% of those losses. It is valid however to use the total loss figure, and simply adjust the R and C values to compensate for this.

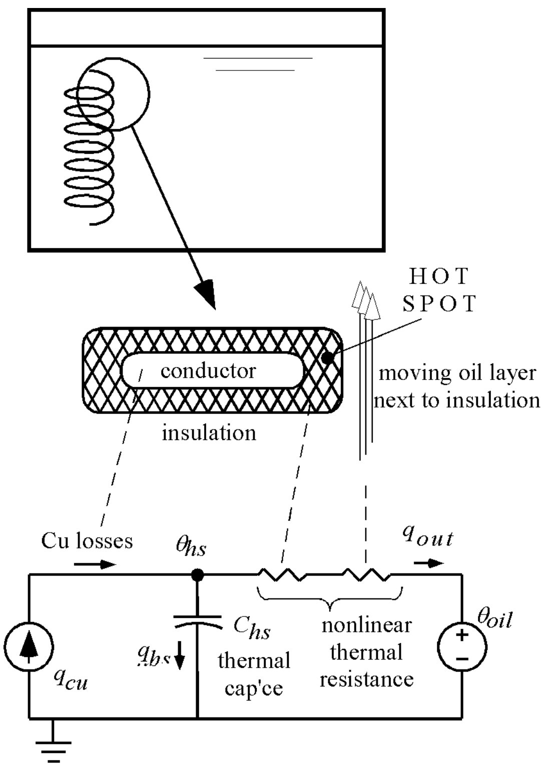

Fig. 2. Thermal model for winding-to-oil heat transfer.

Traditionally, the nonlinearity exponent for this case is given the symbol m.

The physical situation and corresponding thermal equivalent circuit is shown in Fig. 2. The equations are not presented here. They are completely analogous to those for the oil temperature model. Also, physical verification of the *winding-to-oil* or *hot spot* model is not included here: it would involve the installation of fiber optic sensors in a transformer, which is left as a future endeavor.

D. Overall Model

Fig. 3 shows the final overall model. It has certain features:

1) It uses the "current source" concept as the copper loss (and core loss) heat source.
2) It uses the "voltage source" concept as the ambient air heat source (or sink), and as the "ambient" oil heat source (or sink).
3) The nonlinearities are included in an easily understood way.
4) It is not a complicated equivalent circuit.
5) It has been verified physically, as discussed in the companion paper.

IV. CONCLUSION

A thermal model of a power transformer in the form of an equivalent circuit, based on fundamental heat transfer theory, is shown to be a sound basis for the differential equation used to calculate the top oil temperature in large power transformers.

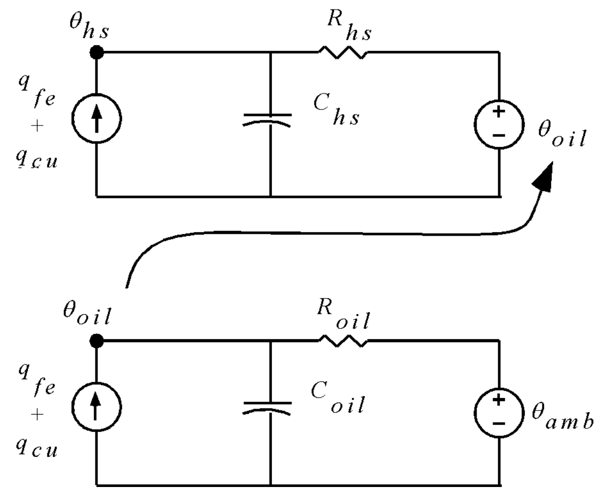

Fig. 3. Overall circuit model.

An analogous thermal model and equivalent circuit for hot spot temperature determination is also presented.

Verification of the top oil temperature model is shown in a companion paper [15].

APPENDIX A
COMPARISON WITH THE C57.91-1995 EQUATIONS

The differential equation (6) governing the (top) oil temperature rise, as derived in this paper is repeated here:

$$\frac{I_{pu}^2\beta+1}{\beta+1}\cdot[\Delta\theta_{oilR}]^{1/n}=\tau_{oil}\frac{d\theta_{oil}}{dt}+[\theta_{oil}-\theta_{amb}]^{1/n}.$$

A differential equation can be inferred from the "exponential response" quoted in the overload Guide[3]:

$$\left[\frac{K^2R+1}{R+1}\right]^n\cdot\Delta\theta_{TO,R}=\tau_{TO,R}\cdot\frac{d\Delta\theta_{TO}}{dt}+\Delta\theta_{TO} \quad (8)$$

where

K = I_{pu} of this paper (the symbol K is not conventionally used to represent a *variable*),
R = β of this paper (the symbol R is conventionally used only for resistances),
$\Delta\theta_{TO,R}$ = top oil rise under rated conditions,
$\tau_{TO,R}$ = τ_{oil} of this paper, and
$\Delta\Theta_{TO}$ = top oil rise, the solution variable.

There are several significant differences between (6) and (8):

1) The dependent variable in (8) is the top oil temperature rise whereas it should be the top oil temperature itself, as in (6).
2) Ambient temperature variation is not included in (8) whereas it is in (6).
3) The exponent n appears in the wrong place, in (8).

Nevertheless, (8) turns out to be adequate as long as the ambient temperature is constant, and as long as the parameter n is not too different from unity. With regard to the latter, a long-standing "correction factor" applied to the time constant has been used [13]. This does indeed make the solution better, but not properly accurate, since the problem is not with the time constant itself but with the shape of the solution curve.

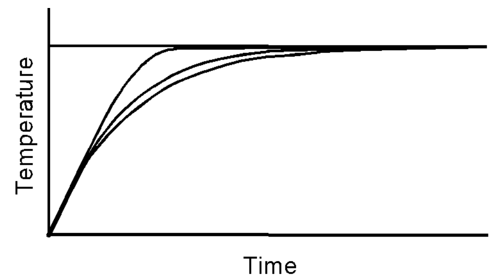

Fig. 4. Step response for $n = 1$ (bottom curve), $n = 0.8$ (middle curve), and $n = 0.3$ (top curve).

To illustrate, Fig. 4 shows the response to a step change in load current for $n = 1$, $n = 0.8$, and $n = 0.3$. A value of $n = 0.3$ does not occur in practice, but is used to illustrate that true "exponential response" does not happen if n is not equal to one. These plots are solutions of (6), with for step change in I_{pu}.

APPENDIX B
HISTORICAL NOTE

There is a very interesting history regarding (5).

If one does a literature search on the origins of n one is led through papers in the 1990s, the 1980s etc. back through the power-related transactions. Finally, one is led to an interesting paper in the American Institute of Electrical Engineers (AIEE) Transactions of 1916 [14]. On page 604 of that paper, it is stated that

> "In 1817, Dulong and Petit announced the following law as a result of their experiments conducted over a rather limited range of temperatures: *The velocity of the cooling due solely to the contact of a gas is proportional to the excess of temperature in degrees centigrade raised to the power 1.223. (italics added here, for emphasis)* This was later verified by Peclet."

It is fascinating that the reciprocal of 1.223 is approximately 0.8, a value of n commonly used to this day, for natural cooling conditions! Incidentally, there is no indication of how this result was "announced," or any reference identifying "Peclet."

ACKNOWLEDGMENT

Valuable input from Prof. G. Sims of the Department of Mechanical Engineering at the University of Manitoba is greatly appreciated. Written exchanges with L. Pierce have also been very helpful.

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